Question 1 - Ethics, Law and Professional Responsibility in Software Engineering.

a) Ethics, Law and Professional Responsibility

Ethics :- Principles of right and wrong behavior.

Law :- Rules made and enforced by the government.

Professional responsibility :- Duties of professionals ~~twords~~ toward users employers and society.

In health app case, using health location data for advertising may be legal in some situations, but it can still be unethical if user did not expect it.

A professional engineer protect privacy and recomend obtaining proper consent

b) Historical Ethical Failure :-

Volkswagen emissions scandal :- Sofware was used to manipulate emissions tests. Customers, regulators and society were affected. Software must not be used for deception.

Boeing 737 max :-

Problems in automated flight-control software and engineering processes contributed to fatal accidents. Safety ~~mst~~ must be important than cost or deadline.

c) Software Engineering Ethics:-

Major principles are :-

- Protect public safety
- Respect privacy
- Maintain ~~seq~~ security
- Be honest
- Work competently
- Avoid discrimination
- Protect confidentiality
- Consider user's interests.

d) The code guides engineers when business interests conflict with public interest. It emphasizes public safety, privacy, honesty, fairness and responsibility. Engineers should raise concern and avoid releasing

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systems that could seriously harm in users.

Question 2 :- Ethical framework and privacy :

a) Utiliterianism vs Deontology.

Utiliterianism judges actions by their consiquences. Data collection may be acceptible if benefits are greater than harms.

Deontology focuses on duties and rights. Secretly analyzing private messages is wrong if it violates user's privacy even if it increases company profits.

b) Virtue Ethics :- Virtue ethics focuses on the character of the person making the decision.

A good software engineer should demonstrate:

- Honesty – Clearly explain data use.
- Fairness – Treat users equally.
- Responsibility :- Consider possible harm.
- Courage :- Report unethical practices.
- Respect :- Protect user's privacy.

c) Ethical Decision Making Flow:-

1. Identify the ethical issue.
2. Identify stakeholders.
3. Apply ethical principles.
4. Check legal and professional duties.
5. Consider alternatives.
6. Select the most ethical options.

d) Legal but Unethical:-

An activity may be legal but still unethical.
For example, a company may technically
disclose data collection in a complicated
privacy policy. But users may not
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also consider fairness, transparency, trust and user expectations.

Question 3 :- Explainability and Black-Box AI.

a) Importance of transparency :-

Transperancy means providing information about how an AI system works and uses data. Explainability means providing understandable ~~reasons~~ reasons for individual decisions.

These are important in lending, healthcare, enjoyment and criminal justice because automated decisions can seriously affect peoples lives.

b) Two major Societal harms :-

Two major harms are :-

1. Discrimination :- The AI may produce unfair outcomes for certain communities.

2. Lack of accountability :- It becomes difficult to determine why an incorrect decision occured or who is responsible.

Other harms include loss of trust and inability to correct mistakes.

c) Ethics of Lacks of Explainability :-

Without explaination, a person may not know why a loan was rejected. They cannot easily correct wrong information or challenge the decisions. Organizations may also find it difficult to detect discrimination.

Therefore, explainability support procedural fairness, accountability and public trust.

d) Information Provided:

An affected person should ideally recieve:

- Information that AI was used
- Main reasons for the decisions.

- Important factor considered.
- Information about possible errors.
- A way to request human review.
- A method to appeal or challenge the decision.

The explanation should be understandable and useful rather than simply providing technical details.

Question 4:- ACM code and professional decision making:

a) ACM Principles:-

Important ACM principles include:-

- Contribute to society and ~~hu~~ human well-being.
- Avoid harm
- Be honest and trustworthy
- Be fair
- Respect privacy
- Honor confidentially
- Maintain professional competence.

Professional codes provide guidance when professions face difficult ~~decis~~ decisions.

b) Relevant Principles :-

The engineer knows that serious security weaknesses exist. Avoid harm, honesty, privacy, security and professional competence are directly relevant.

The engineer should not hide the weakness simply to meet a deadline.

c) Professional Responsibility :-

The engineer should document the security problems, inform manager and recommend delaying the release until serious ~~w~~ weakness are fixed. If management refuses, the matter should be ~~es escale~~ escaled through appropriated channels.

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d) Code and Judgement :-

- A professional code provides principle but does not automatically give one answer for every situation. Engineers must use professional judgement based on facts, risks, laws and consequences.

Question 5:- Whistleblowing and Professional Dissent..

a) Definitions :-

Professional dissent means respectfully disag disagreeing with a decision because of ethics, safety or technical concern.

Whistleblowing means reporting serious wrongdoing to an appropriate authority, sometimes outside the organization when internal reporting fails.

b) Ethical Consideration :-

The analyst should :

- verify the vulnerability.

- Preserve evidance
- Assess possible harm
- Report internally first.
- Give management an oppotunity to fix it.
- Consider external disclosure only if serious risk remains.

c) Risks:-

The whistleblower may face job loss. profession damage, legal problem, retailiation, and personal state.

However, keeping a serious vulerability secret

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may expose many users to harm.

d) Decision Path :-

Discover Wrongdoing → Verify Evidance.

↓

Report Internally ← Assess harm

↓

Evaluate Response → Consider External Disclosure.

↓

Minimize unnecessary disclosure

Each steps helps prevent false accusations and unnecessary damage.

Question 6 :- Privacy vs Social Benefit.

a)

Using mobile location data ~~dri~~ during a disease outbreak may help authorities identify infection patterns and saves lives. This supports the principle of contributing to society.

However, location information can reveal sensitive personal activities. So the Principle of respecting privacy must also be considered.

b)

Important principles are :

- Informed Consent
- Purpose limitation
- Data minimization
- Security
- Transparency
- Accountability.

Data should be collected only for a clear purpose and protected from unauthorized access.

© Researcher should use the minimum information needed prefer agregated or

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anonymized data where possible. They should compare the expected public heath benefits with privacy risks and choose the least intrusive method.

d) Safeguards include encryption, access controls, anonymization, limited retention, secure deletion, audits, ethical review, and strict limit on data sharing.

Question 7:- Copyright, Pattents and Trademarks:

a) Differences.

Copyright protects original creative works such as source code, documentation and graphics.

Patents ~~ponet~~ protects qualifying inventions and technical innovations.

Trademarks protects names, symbol, logo, and other identifiers that distinguish a company's products.

b) Appropriate Protection :-

The startup's original code should mainly be protected by copyright.

A new hardware-assisted algorithm may potentially qualify for patent protection, depending on applicable law.

The unique products name and logo should be protected using trademark protection.

c) Importance and Ethical Concerns :-

IP ~~rol~~ protection encourages innovation by giving creator control over their work.

12 min.

However, excessive IP restrictions may make education, research, interoperability and innovation more difficult.

Therefore, intellectual-property rights should be respected while also considering broader social benefits.

d) Improper Copying :

Copying another company's source code without permission can violate copyright or license condition. It may result in so lawsuits, financial penalties,

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loss of reputationy, removal of the software and professional disciplinary action.

<u>Question 8</u> :- Open Source proprietary Software.

a) Comparison

Open source software allows users to use, inspect, modify and redistribute software according to its license.

Proprietary software is controlled by its owner, and users recieve limited rights under a licence agreement.

b) Advantage and Concern

Open source provide transparency, collaboration accessibility and research oppotunities.

Propritary software provides commercial control, business protection. ~~ad~~ and predictable ownership.

Both models have ethical responsibilities, especially regarding security, attribution and ~~itt~~ intellectual property.

c) Third party licenses

Developers must check licenses because they may require attribution, sharing of modifications or other conditions.

Ignoring license terms can create legal and professional problems.

d) Licensing Decision

The university should consider:

- Security
- Openness
- Accessibility
- Commercial goal
- Intellectual-property rights
- Sustainability
- Community contribution

Question 9 : Responsible Disclosure

a) Definition

Responsible disclosure means reporting a cybersecurity vulnerability to the responsible organization so that it can investigate and fix the problem.

It differs from cybercrime because the researchers avoids unauthorized exploitation, unnecessary access and damage.

b) Guidelines :-

The researcher should :

1. Confirm the vulnerability safety
2. Avoid accessing unnecessary personal information

3. Document the finding

4. Report securely

5. Allow reasonable time for fixing.

6. Coordinate public disclosure.

e) Avoiding user data

Accessing real user data simply to prove a vulnerability creates additional privacy and legal risks. Test accounts or limited ~~demos~~ demonstrations should be used whenever possible.

d) Flow

Discover → Confirm safety → Avoid harm

↓

Allow Remediation ← Report ← Identify organization

↓

Coordinate disclosure

The purpose is to reduce risk and protect users.

Question 10 : Ethical Hacking vs Cybercrime

a) Difference

Ethical hacking is authorized testing performed to improve security.

Cybercrime involves unauthorized or illegal activities such as theft, damage or unauthorized access.

b) Condition for Ethical Hacking

Ethical hacking requires:

- Proper authorization
- Clearly defined scope
- Legitimate purpose

- proportional testing
- Privacy protection
- Avoidance of unnecessary damage.

c) Scope Violations

Accessing student records, attacking systems outside the agreed scope or disrupting university services. ~~who~~ would be unethical and unauthorized.

d) The Technical Ability

Technical skill does not create permission. An activity becomes ethical only when

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it has proper authorization, purpose scope, and responsible execution.

Question 11 : Algorithmic Bias in Recruitment

a) Definition and Causes

Algorithmic bias occurs when an AI system producess systematically unfair results.

causes include :

1. Biased training data

2. Historical discrimination

3. Proxy variables.

4. Poor ~~s a~~ sampling

5. Incorrect objectives

b) Detection

Bias can be measured by comparing selection rates, false positive rates, false negative rates and accuracy across demographic group.

c) Improvements :-

Organization Should :-

- Improve training data
- Test fairness
- Audit models
- Use diverse development teams
- Provide human review
- Monitor the system continuously

d) Removing Demographic Variable

Removing gender or ethnicity may not eliminate bias because other variables such as location, education or employment history may act as proxy variables. Therefore, the whole system must be evaluated for fairness.

Question 12 :- Accountability in Autonomous System

a) Meaning

Accountability means being responsible for decisions, actions, and consequences.

Although an AI autonomous system makes decisions Automatically, people remain responsible for designing testing and develop it.

b) Responsibilities

- Developers : Correct and safe Design.

- Testers: Proper Testing
- Managers:- Responsible deployment decisions
- Manufacturers: Product Safety.
- Operators: Correct use
- Organization: Monitoring and remediation.

c) "AI Made the Decision"

AI cannot simply be blamed for harm. Humans selected the data model, objectives and operating conditions.

8 min

therefore, saying "The AI Made The Decision" does not remove human accountability.

2) Accountable Chain

Requirements → Data → Model → Implement

↓

Implementation → Testing → Deployment

↓

Monitoring → Incident Response →

↓

Remediation

Failure at any stage can contribute to harm.

Question 13 : Ethical Leadership in IT.

a) Meaning

Ethical leadership means creating an organization culture based on honesty, fairness, safety, responsibility and respect.

b) Creating Ethical Culture

Leader should provide :

- Clear ethical policies
- Training
- Open communications

- Fair incentives
- Tranparency
- Accountability
- Protection for employees who ~~report~~ genuine concerns.

© Performance Target

If employees are rewarded only for fast product releases they may ignore security, privacy or g quality problems

Therefore, performance measures should also include security, reliability.

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quality and ethical behavior

4) Reporting framework

Report → Document → Investigate → Escalate
↓
Correct problem ← Protect employee

Employees should be able to report concerns without fear of retaliation.

Question 14 : Informed Consent

a) Meaning

Informed consent means that user understands what information is collected, why it is collected and how it will be used.

b) Long Privacy Policy :-

Clicking "Accept" on a lengthy policy does not always mean users have meaningful understanding. Important information should be presented clearly.

c) Characteristics :-

Ethical consent should be :

- Clear
- Transparent
- Voluntary
- Specific
- Understandable.

d) Better process :-

The application should clearly tell users that precise location is continuously collected and explain why, who recieves it and how long it is stored.

Users should have meaningful choice for optional uses such as advertising.

Consent should be understandable, not merely a formal acceptance.

Question 15 :- Surveillance Technology

a) Ethical Concerns :-

Major concerns include :

- Privacy loss
- Lack of consent
- Misuse
- Discrimination
- Security Risks

- Excessive monitoring
- Lack of oversight

6) Benefit and Risks :-

Facial Recognition may help identify the criminal but can produce false mathes

Location tracking can support emergency services but reveal personal movements.

Workplace monitoring may improve ~~Produ~~ productivity but can create stress and excessive survalliance.

c) Misuse :-

Without safeguards, survallance information may be shared, misused or retained for too long.

d) framework :-

Need → Necessity → Proportionality

↓

Security ← Legal Authority

↓

Oversight → Limited retention

↓

Review

Surveillance should be necessary.

proportionate and properly controlled.

Question 16 :- Copyright and Patents :

a) Comparison

Copyright protects original expression, while patents protects qualifying technical inventions.

b) Protection :

Source → Code → Copyright

GUI → copyright

Image-compression invention → Possible Patents

Company name → Trademark

c) Multiple Protection :-

One software product can contain different forms of ~~intte~~ intellectual property, so several protection may be required.

d) Ethical Responsibility :-

Developer must not copy protected code documentation or designs without permission

Respecting IP is both a legal and professional responsibility.

Question 17 :- Digital Divide.

a) Defination :-

The digital divide is the gap between people who can access and effectively use technology and those who cannot.

It include differences in :-

- Access
- Skills
- Quality
- Effective use

b)

b) Effects :-

It can affect :

- Education
- Employment
- Healthcare
- Government Services
- Economic oppotunities.

c) Solutions

Solution includes :-

- Affordable internet
- Rural Infrastructure
- Low cost devices

- Digita literacy program
- Accessable websites
- Community technology centers.

d) Connectivity Alone :-

Internet access alone is insufficient. People also need suitable devices, skills, affordability and accessable services.

Digital inclusion requires both access and the ability to use the technology effectively

18) Question 18 :- Green Computing.

a) Definition

Green computing means using technology in ways that reduce energy consumption, pollution and electronic waste.

b) Methods :-

Computing professionals can use

- Energy-effective hardware
- Virtualization
- Efficient software
- Cloud optimization
- Equipment reuse.

- Responsible recycling.

c) Software impact :-

Poorly designed software may consume excessive CPU, ~~and~~ memory and network & resources. Efficient algorith can't reduce energy use.

d) Lifecycle :-

Procurement → Efficient operation → Resource Optimization → Reuses → Source data removal → Recycling → Environmental Review

Environmental responsibility should be considered throughout the life of computing Equipments.

Question 15 :- Conflict of interest.

a) Definition

A conflict of interest occurs when personal interests may influence professional judgements.

Here the architect's family relationship whi with the vendors create a conflict

b) Types :-

Actual conflict :- Personal conflict directly affects a decision

Potential conflict :- A conflict may arise later

Perceived conflict :- Other reasonably believed that the decision may be biased.

c) Correct action :- The architect should disclose the relationship to management and, if necessary, withdraw from the final decision.

d) Management :-

Organization should use :-

- Disclosure
- Recusal
- Independent evaluation
- Documentation
- Trasparent criteria.

Transparency protect both the organization and the ~~proffesr~~ professional.

Question 20 :- Privacy, Security and Business Pressure.

a) Issues

The smart-home platform involving :-

- Privacy
- Security
- Unauthorized access
- Personal data
- User safety
- Business pressure
- Professional responsibility

b) Ethical Frameworks

Utiliterianism : Compare business benefits with possible user harm

Deontology :- Engineers have a duty not to knowingly expose users to serious risky.

Virtue Ethics :- A responsible engineer should show honesty courage and responsibility.

c) Professional Principles

Relevant principles include avoiding harm, respecting privacy, honesty, security and public welfare.

d) Recommendation :-

The product should be delayed untill the serious vulnerability is fixed and tested.

e) Dissent :-

Engineers should report first concerns internally. If serious risks remain unsolved, they may escated through appropriate channels and, in ~~ext~~ extreme circumstances, consider whistleblowing.